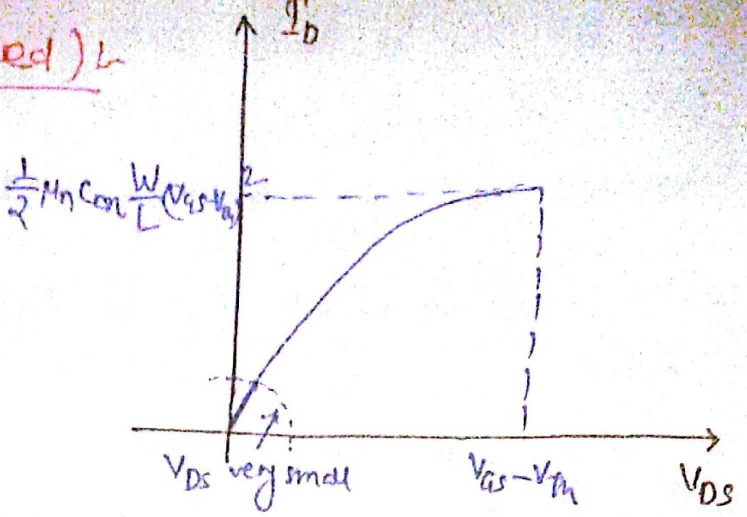
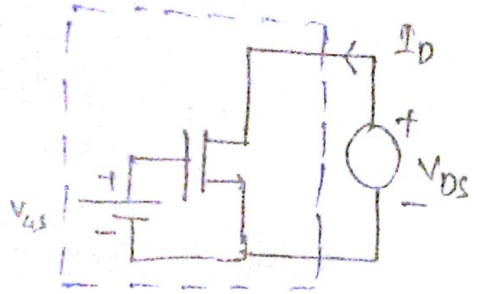
Lecture - 31

* I-V characteristics (continued):-

If $V_{DS} << 2(V_{GS}-V_{TH})$ then

$$\boxed{I_D \approx \mu_n C_{ox} \frac{W}{L} (V_{GS}-V_{TH}) V_{DS}}$$





* V_{DS} is small

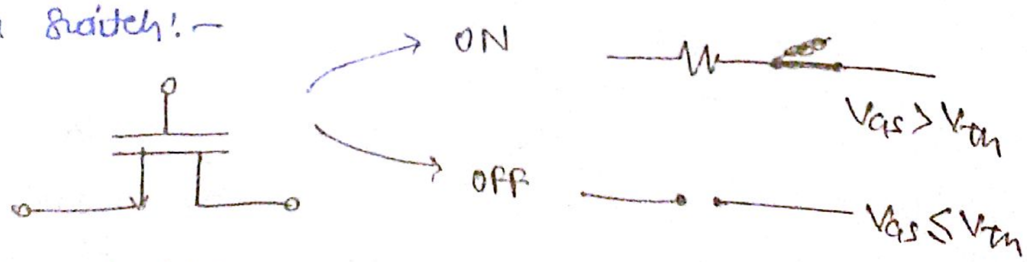
$I = \frac{V}{R}$

$$\Rightarrow \boxed{R_{ON} = \frac{1}{\mu_n C_{ox} \frac{W}{L} (V_{GS}-V_{TH})}}$$

→ This says that MOS device can works as a resistor, when V_{DS} is not too large. Since this resistance value is in our control so it is called as electronically adjustable resistor / electronically tunable resistor.

→ A MOSFET can act as a voltage dependent resistor (if $V_{DS} << V_{GS} - V_{TH}$).

* MOS device as a switch:-

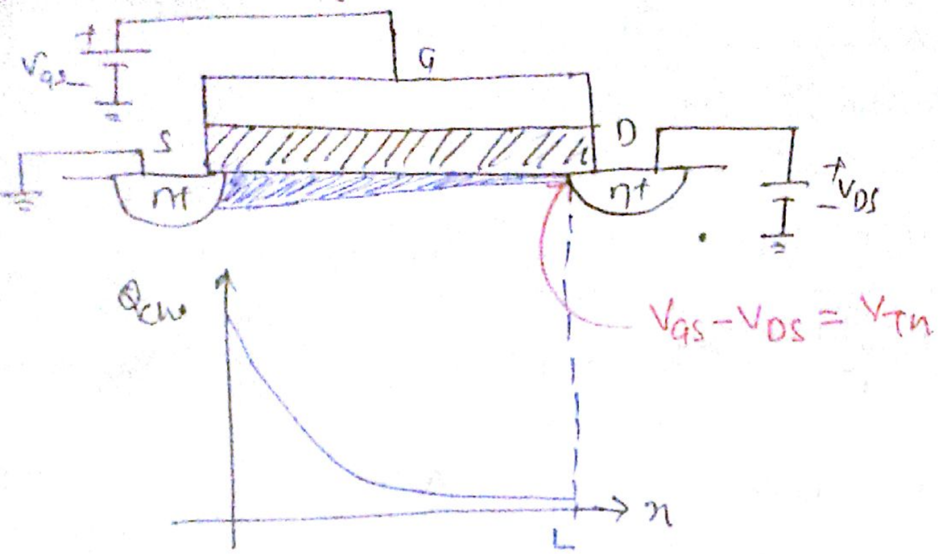


→ If the Gate voltage is high enough then we have a finite resistance between source to drain.

→ If the Gate voltage is low enough then device turns off, we have infinite resistance. So its like a switch but it's not a ideal switch because it has some resistance when it is ON.

* Once the V_{DS} reaches $V_{GS} - V_{TH}$ and increases then the I_D becomes constant.

* I_D for $V_{DS} > V_{GS} - V_{TH}$?

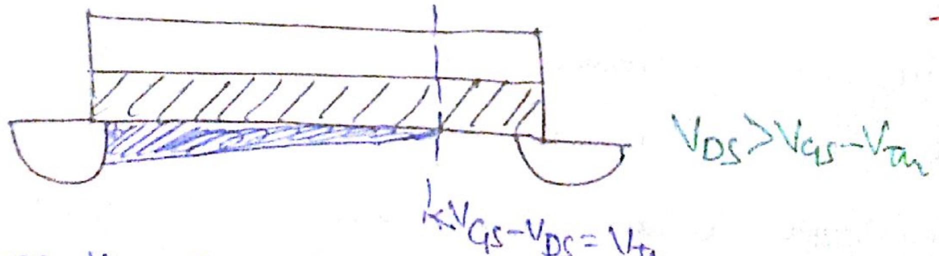


→ The charge density may fall to zero if the voltage difference at drain is not enough to create free electrons at the edge of channel.

* If at the edge $V_{GS} - V_{DS} = V_{TH}$ then there is no charge density

If $V_{GS} - V_{TH} > V_{DS}$ ~~then~~ there is some charge

→ If V_{DS} goes beyond $V_{GS} - V_{TH}$ then point at which the charge density goes to zero is no longer here its a litter farther back. The point at which charge density became zero is called pinch off.



→ As the voltage V_{DS} goes up, this point moves to the left very slowly, its a very weak function. The total channel length does not change much.

To derive the I/V eqⁿ :-

$$\int_0^L I_D\,dx = \mu_n C_{ox} W \int_0^{V_{GS}-V_{TH}} \left[V_{GS} - V_{TH} - V(x)\right] dV$$

$$\boxed{I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH})^2}$$

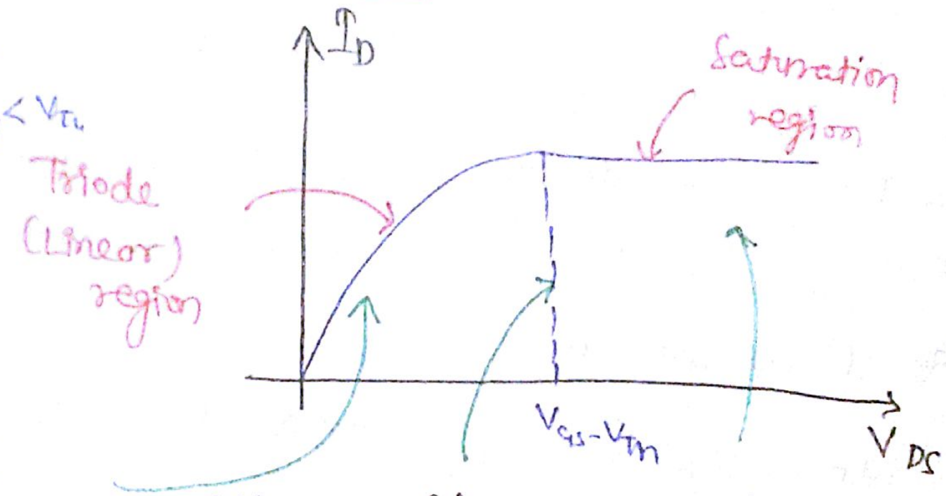
for $V_{DS} \geq V_{GS} - V_{TH}$
Saturation Region

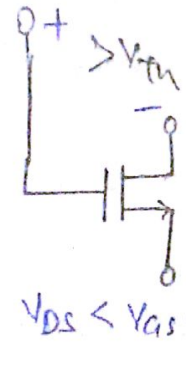
This is equation beyond the pinch off.

$$\boxed{I_D = \mu_n \frac{C_{ox}W}{L}\left[(V_{GS}-V_{TH})V_D - V_{DS}^2\right]}$$

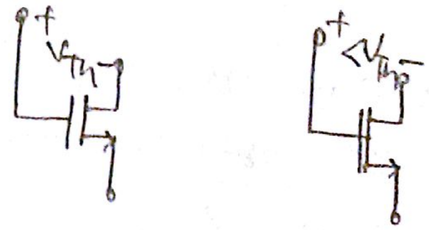
Triode (Linear) region
$V_{DS} \leq V_{GS} - V_{TH}$

→ If the Gate voltage is low enough i.e. $V_{GS} < V_{TH}$ then device is off, i.e., cutoff.





$>V_{TH}$

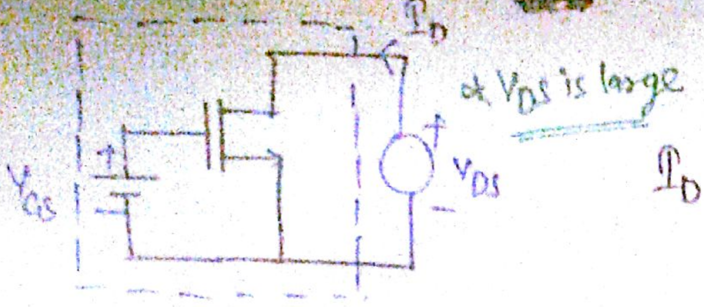


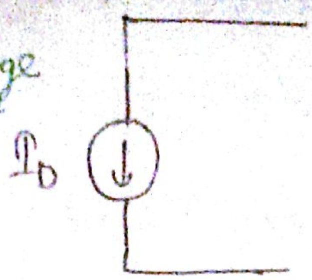
→ If the gate is above the drain by more than one threshold V_{TH} then we are in triode region.

→ If the gate and drain have potential difference V_{TH} then it is at edge.

→ If the gate is below the drain by more than V_{TH} then we are in saturation region.

Observation:-



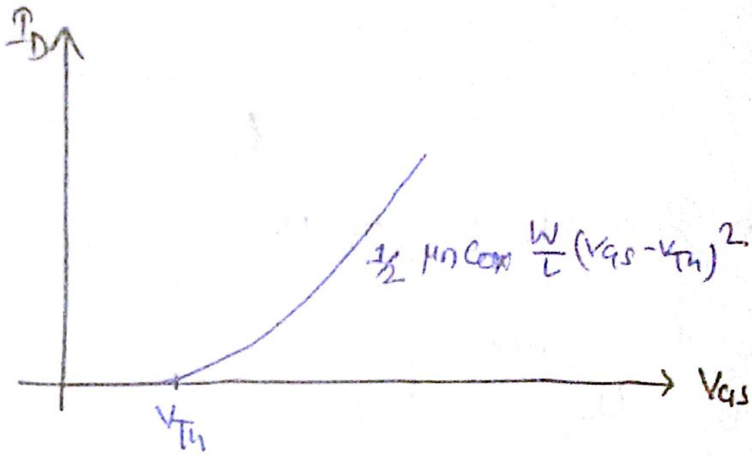


→ If V_{DS} is varying by large amount then current does not vary i.e. Current is constant in device so it works as current source.

* I_D v/s V_{GS} characteristics:-

→ If V_{GS} is below V_{TH}, device is off.

→ If V_{GS} exceeds V_{TH}, we use saturation equation which has square law behaviour.



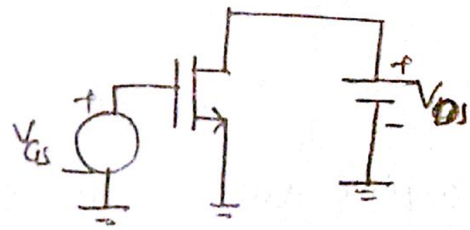
Exersize:- Prove that the MOS device always ON in saturation if $V_{DS}>0$.

Since MOS device becomes ON,

$V_{GS} > V_{TH}$

So $V_{GS}-V_{TH}>0$

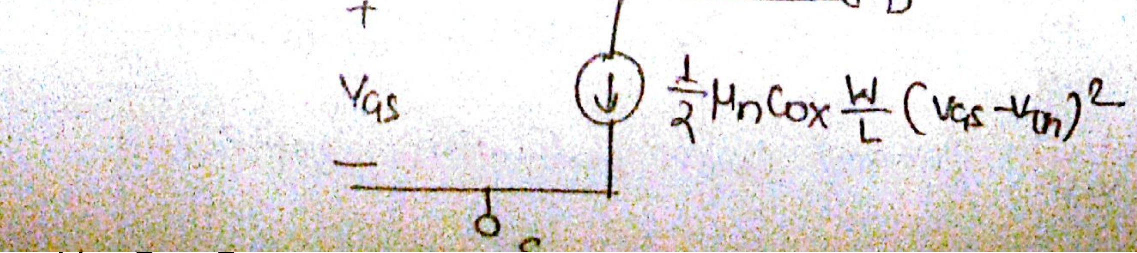
that means $V_{DS} \geq V_{GS}-V_{TH}$



This condition occurs in saturation. Therefore MOS device always ON in saturation if $V_{DS}>0$.

* Simple Model:- In saturation,

voltage dependent current source



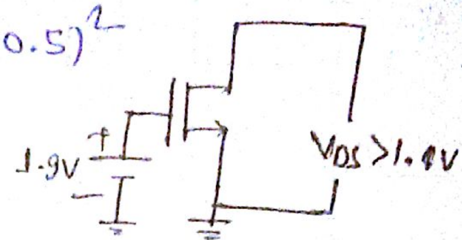
Example: $\mu_n C_{ox} = 100 \mu A/V^2$, $V_{th} = 0.5V$, $\frac{W}{L} = \frac{5 \mu m}{0.5 \mu m}$

Design a 1mA current source.

$1mA = \frac{1}{2} \times 100 \times 10^{-6} \times 10 \times (V_{GS} - 0.5)^2$

$2 = (V_{GS} - 0.5)^2$

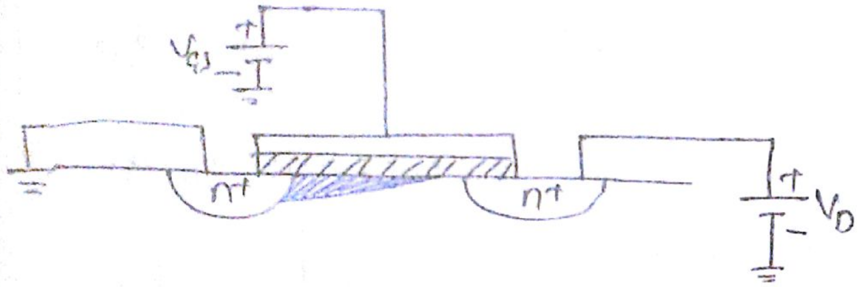
$\boxed{V_{GS} = 1.9V}$

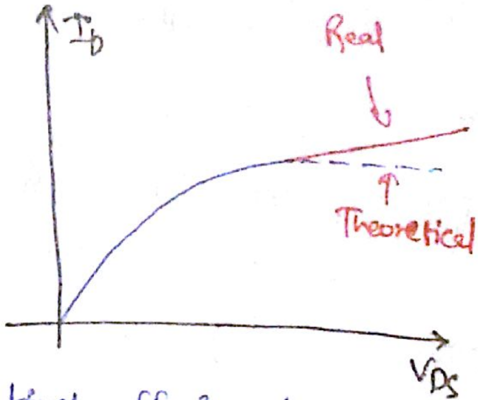


* To act as current source $V_{DS} > 1.4V$.

Lecture - 32

* Channel-Length Modulation (Similar as Early Effect):-





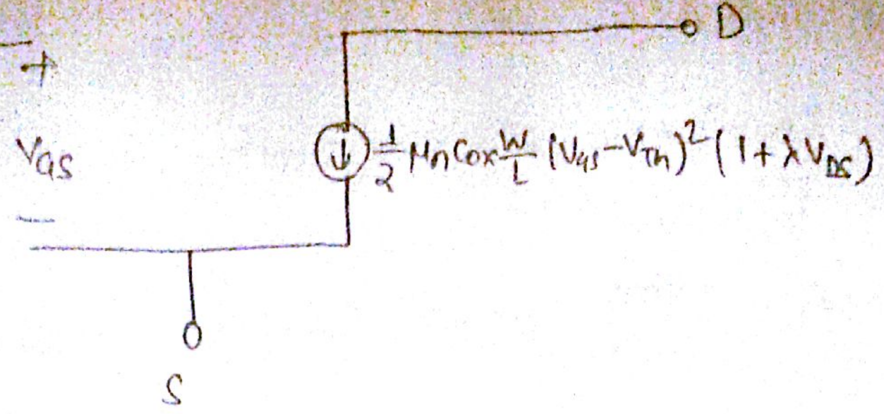
* Suppose we are in saturation so we have pinch off in channel.

→ As $V_{DS} \uparrow$, the channel length L decreases by small amount. Since L is in inversely proportional to I_D. That means I_D depends on V_{DS} also in saturation.

$$\boxed{I_D = \frac{1}{2} C_{ox} \mu_n \frac{W}{L} (V_{GS} - V_{Th})^2 (1 + \lambda \cdot V_{DS})}$$

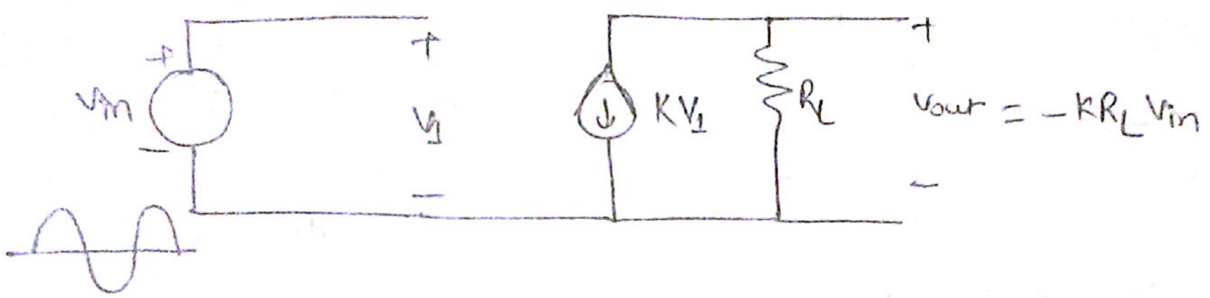
This tells us that as V_{DS} increases, I_D increases.

$\lambda : \frac{1}{Volt}$ Channel length modulation coefficient

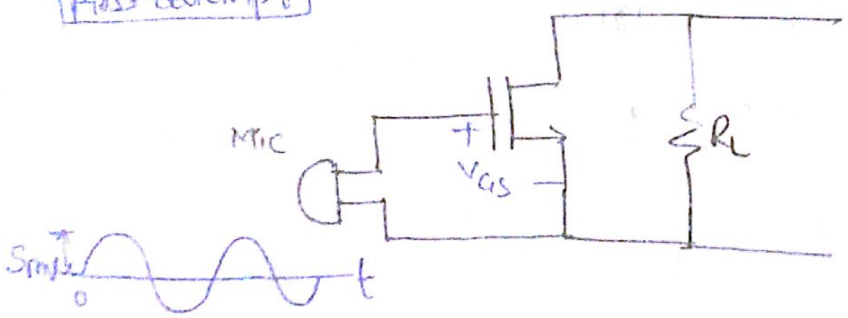


* Complete model of MOS Transistor →

Let's build an amplifier!-



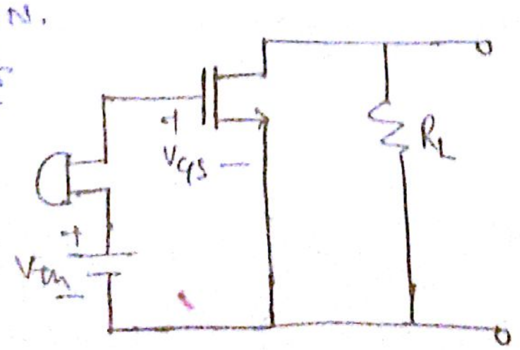
First attempt

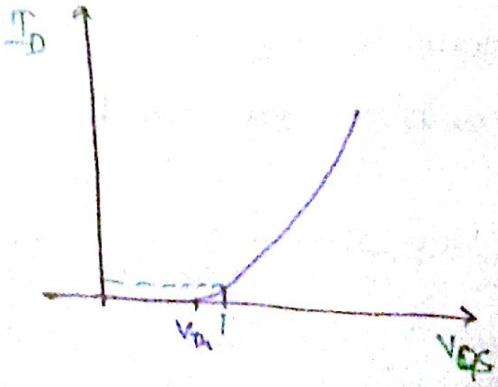


Since device is off untill $V_{GS} \geq V_{TH}$

Since input signal is so small so $V_{GS} < V_{TH}$ therefore device will never turn ON.

Second Attempt





→ If microphone has no signal that means $V_{GS} = V_{TH}$. As the signal comes in $V_{GS} - V_{TH} = 5mV$ so we have some current.

$\mu_n C_{ox} = 100\mu A/V^2$, $\frac{W}{L} = 10$, $V_{Th} = 0.5V$

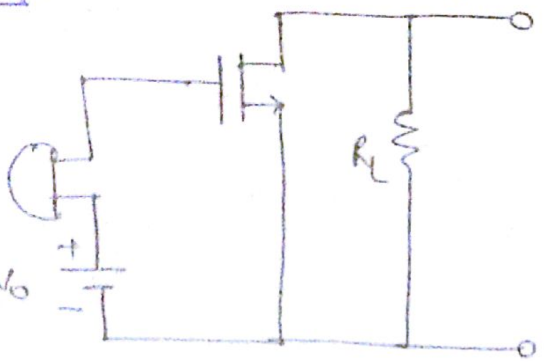
$I_D = \frac{1}{2} \times 100 \times 10^{-6} \times 10 \times (5mV)^2 = 12.5\,nAmp.$

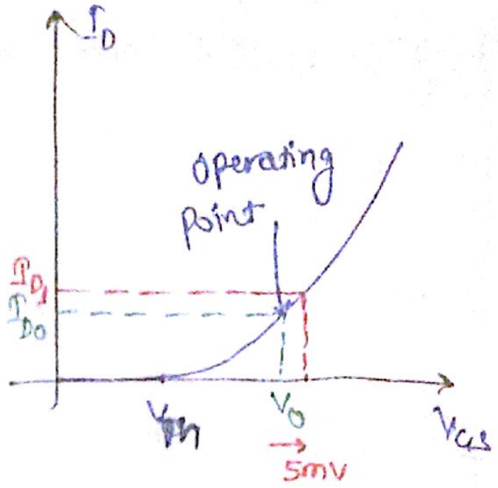
which is so small.

$R_L = \frac{50mV}{12.5nA} = 4M\Omega$ which is very large resistor

Now we move operating point,

Third Attempt





$V_0 = 0.9V$, $V_{Th} = 0.5V$, when no signal

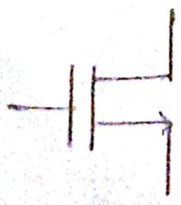
$I_{D0} = \frac{1}{2} \times \mu_n C_{ox} \frac{W}{L} (0.9 - 0.5)^2 \simeq 80\mu A$

$I_{D1} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (0.95 - 0.5)^2 \approx 82\mu A$

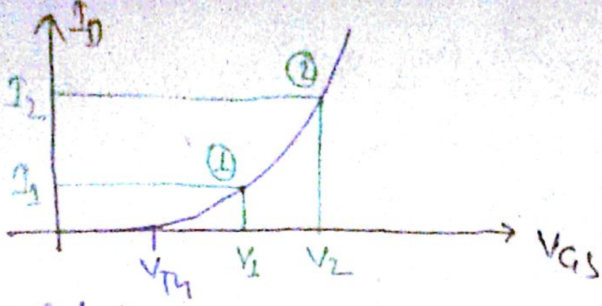
Change in $I_D = 2\mu A$ $\Rightarrow$ $R_L = \frac{50mV}{2\mu A} = 25K\Omega$

* Need to bias the transistor by creating proper current and terminal voltages (in the absense of signal) so that device can amplify the signal.

Observation:- (1) A MOS devices converts a voltage to a current

 $\equiv$ V/I Convertor $\Rightarrow$ Transconductor

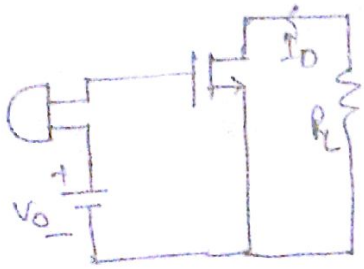
(3) Which operating point is preferred?

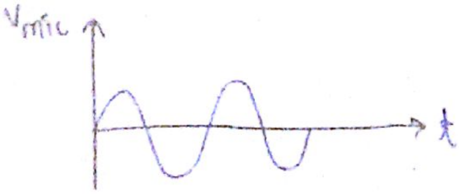


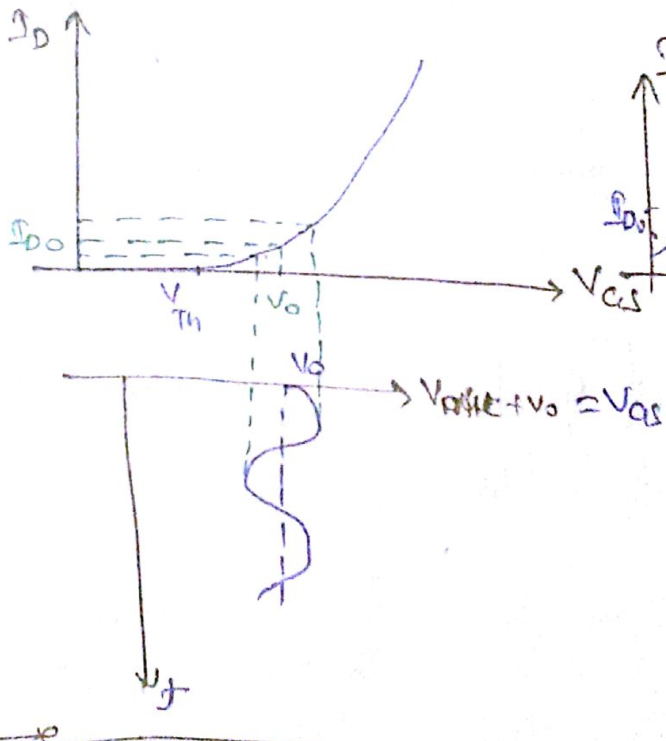
→ For a given input signal change, we get a larger current change at (2) than (1). Transistor is better V/I convertor at (2) than (1). So point (2) will be preferable.

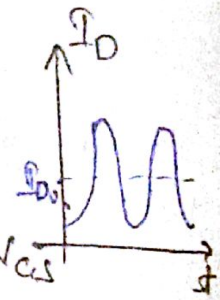
→ When there is no signal, the bias current is draining the battery. If we want more stronger device, so the drain current drains the battery more quickly.

* Combining Time response with I/V characteristics :-



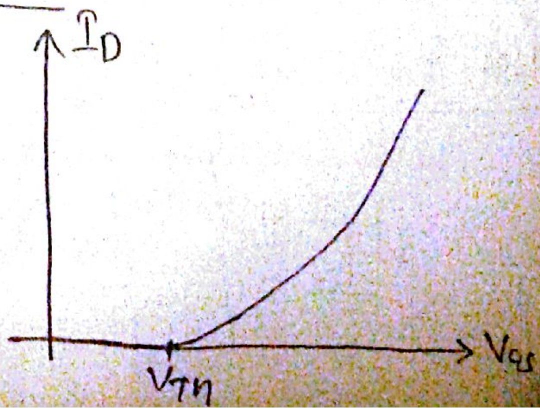






Concept of Transconductance :-

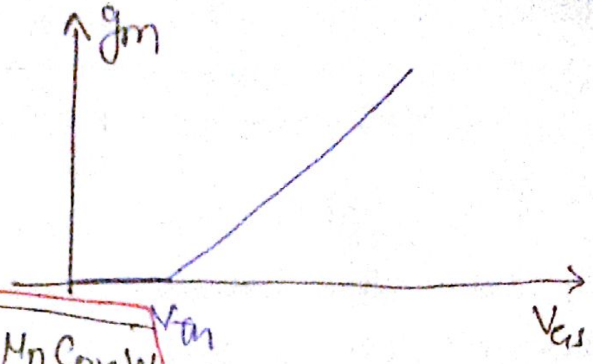
$$g_m = \frac{dI_D}{dV_{GS}} \quad \frac{1}{\Omega} \text{ or Siemens } (S)$$



→ The g_m tells us that how stronger is transistor.

$$\therefore I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{Th})^2 \quad , \lambda=0$$

$$\boxed{g_m = \mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{Th})}$$

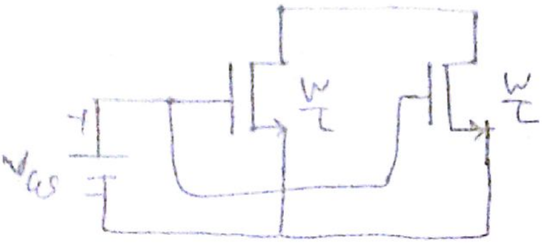


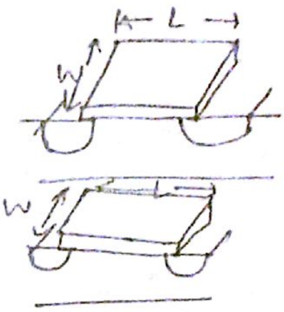
$$\boxed{g_m = \frac{2I_D}{V_{GS}-V_{Th}}}$$

$$\boxed{g_m = \sqrt{2I_D\,\mu_n C_{ox}\frac{W}{L}}}$$

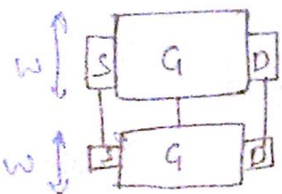
Lecture - 33

Example:-





When another MOS is connected in parallel with a MOS then the combination have $\frac{2W}{L}$.



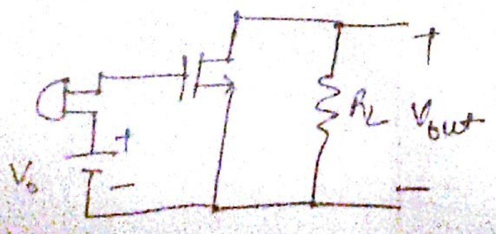
$$I_{D_1} = I_{D_2} = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{Th})^2$$

$$I_{D_1}+I_{D_2} = \frac{1}{2}\mu_n C_{ox}\left(\frac{2W}{L}\right)(V_{GS}-V_{Th})^2$$

Since both has same V_{GS} & V_{Th}, μ_n, C_{ox}.

$$\Rightarrow g_m = \frac{2I_{D_1}}{V_{GS}-V_{Th}} \Rightarrow \text{New } g_m = 2\cdot\frac{2I_{D_1}}{V_{GS}-V_{Th}}$$

Let build an amplifier:-



* We do need to make sure device is in saturation.

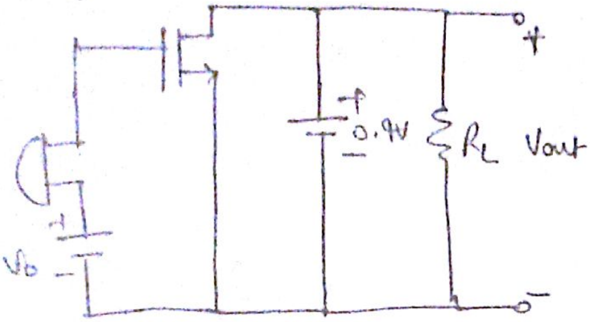
$$V_{DS} > V_{GS}-V_{Th}$$

Say $V_0 = 0.9V$, $V_{TH} = 0.5V$

For device in saturation,

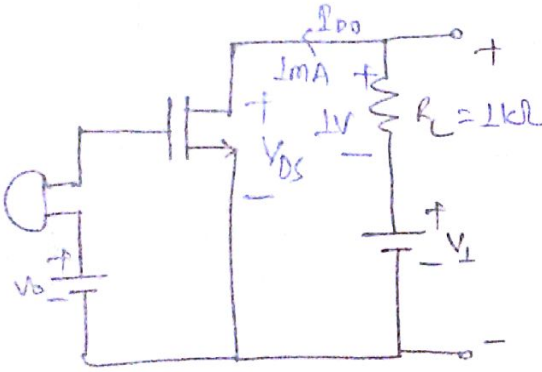
$$V_{DS} \geq (0.9 - 0.5) \Rightarrow V_{DS} \geq 0.4V$$

Since a dc current can not flow through circuit until it is applied with a source. Therefore V_{DS} would be zero.



Is this device in saturation?

Yes, it will be in saturation but we don't have Vout we have just 0.4 across the R_L so we don't have output voltage varying time.



$$V_1 = V_{DS} + I_{D0} R_L$$

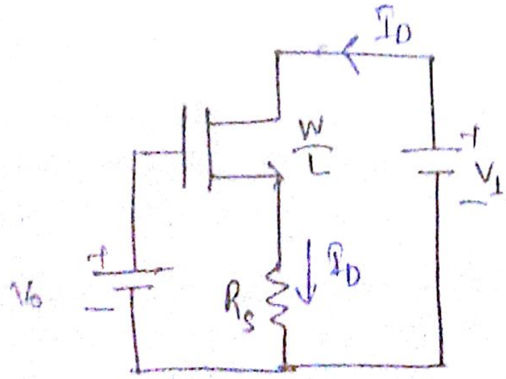
$$\Rightarrow V_{DS} = V_1 - \underbrace{I_{D0} R_L}_{1V}$$

$$0.4 = V_1 - 1$$

$$\Rightarrow \boxed{V_1 = 1.4V}$$

So for device in saturation we should choose $V_1 = 1.4V$.

Large Signal Operation:-

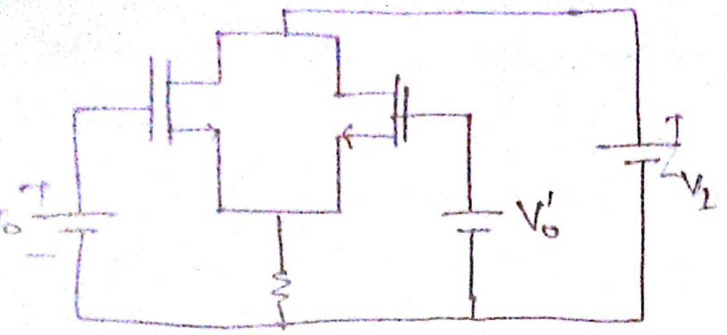


$$V_0 = V_{GS} + I_D R_S$$

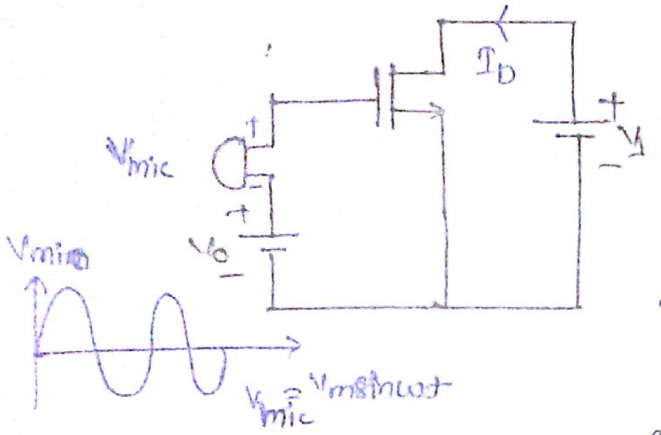
$$\because V_{GS} = \sqrt{\frac{2 I_D}{\mu_n C_{ox} \frac{W}{L}}} + V_{TH}$$

$$\Rightarrow V_0 = \sqrt{\frac{2 I_D}{\mu_n C_{ox} \frac{W}{L}}} + V_{TH} + I_D R_S$$

Previous results would not help if we have another types of ckt.



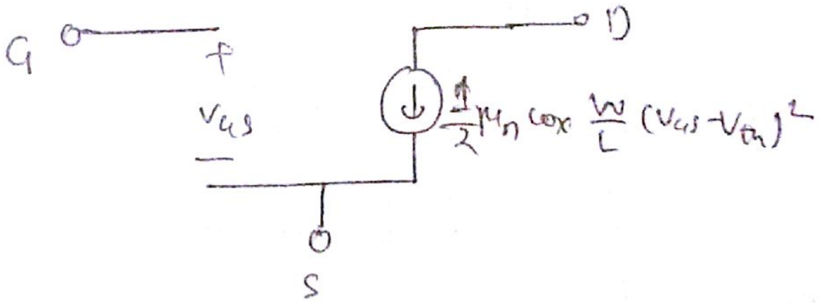
Let's add a signal :-



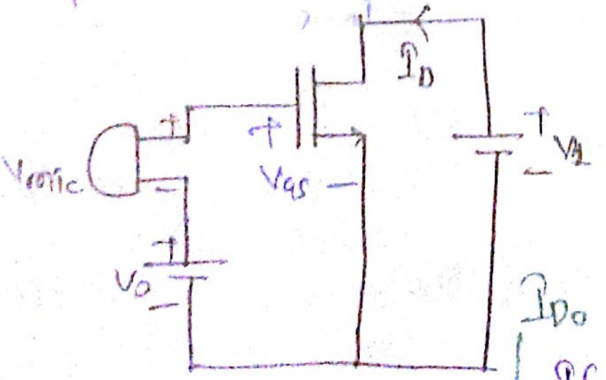
$$V_o + V_m \sin\omega t = \sqrt{\frac{2I_D}{\mu_n C_{ox}\frac{W}{L}}} + V_{Th} + I_D R_s$$

This is called large signal operation, if V_m is not small.

⇒ Use large signal model of Transistor:



★ Small signal operation:- In this signal comes in is small in amplitude.



$$V_{GS} = V_o + V_{mic}$$

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{Th})^2$$

$$= \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_o + V_m \sin\omega t - V_{Th})^2$$

If V_m is small, $V_m << V_{GS} - V_{Th}$

$$\boxed{I_D = \underbrace{\frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{Th})^2}_{I_{Do}}\left(1+\frac{V_m \sin\omega t}{V_{GS}-V_{Th}}\right)}$$

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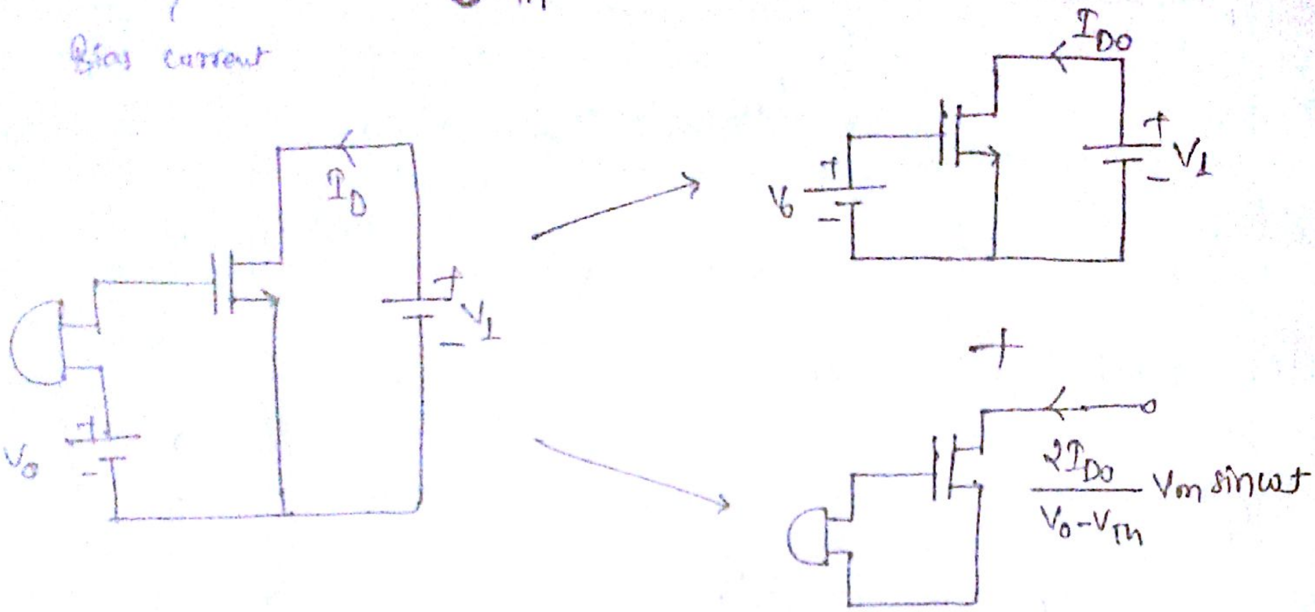
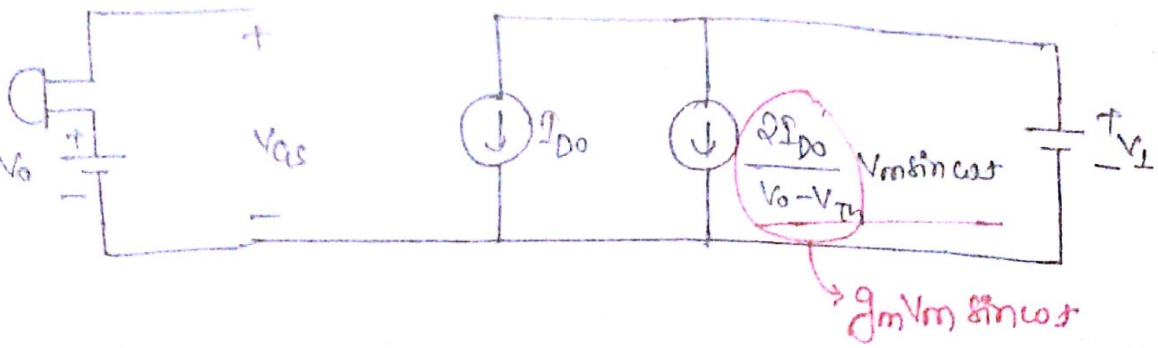
Because $(1+\epsilon)^2 \approx 1+2\epsilon$

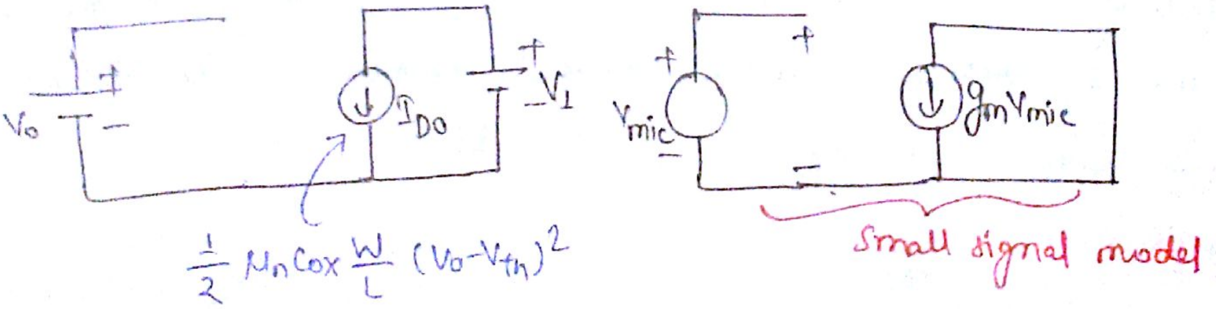
Signal current

$$I_D = I_{D0} + 2\,\frac{I_{D0}}{V_{GS}-V_{Th}}\,V_m \sin\omega t$$

Bias current







$\frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_0-V_{th})^2$

Small signal model

* The small signal consists of something that represents only changes in the currents and voltages.
* Everything which does not change with time, it has zero value in small signal model.

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